Flying Wing VTOL 4+1 Drone

Presented by

RAHIB Achraf

Project Flying Wing QuadPlane Components Overview

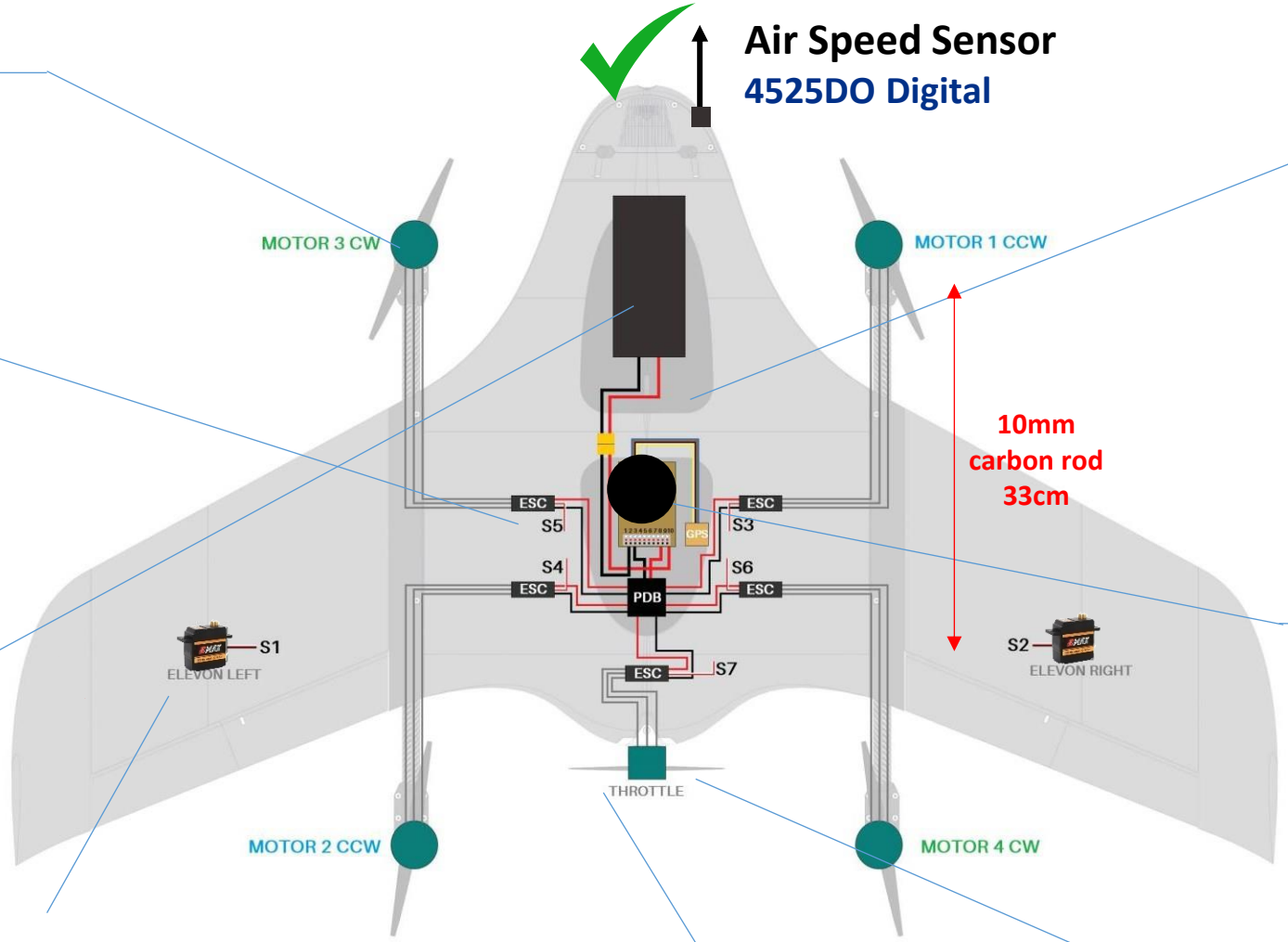
VTOL motors:
T-Motor AT8020
With 25x10 props

ESC:
- 4x JetiModel ESC
Mezon Evo 85 opto

- 1x MasterSPIN
165 Pro OPTO

Power Source VTOL:
2pcs 6000 mAh
Batteries LIPO for
left/right VTOL engines

Servos: 2 servos 75 kg .
For Elevons
+ 900 mAh 2s battery



Air Speed Sensor
4525DO Digital

GPS + Compass
Holybro M9N GPS

Radio control:
Radiomaster Pochet
ELRS Radio

Flight Controller
Pixhawk 6c running
Arduplane

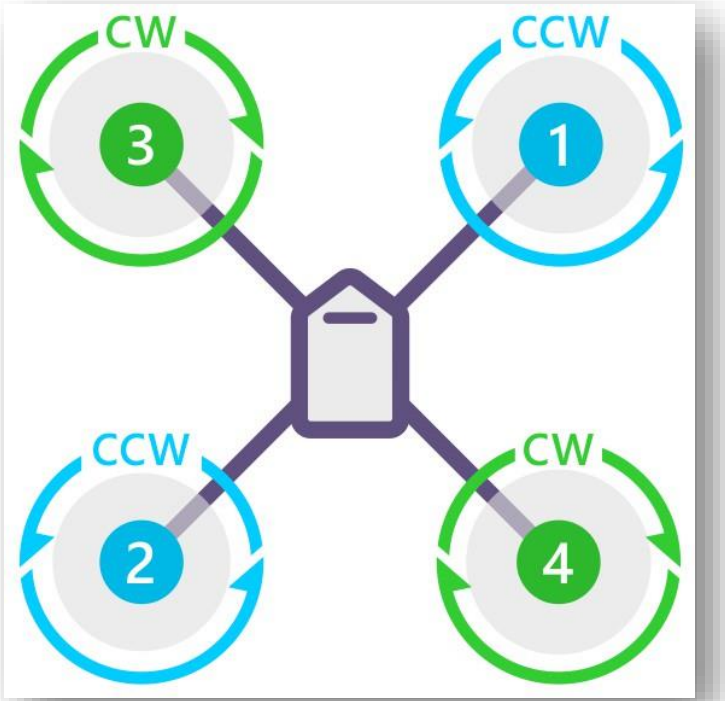
Propellers:
23x12

Power Source Pusher:
2 pcs 7500 mAh batteries

Project Flying Wing QuadPlane Components Overview

X Frame Configuration

The configuration used is a classic X Frame layout. Two pairs of propellers are needed: one pair of CW and one pair of CCW, as indicated in the drawing.



For example, with the default Quad-X frame the motors are on outputs 5 to 8. The arrangement is:

Output 5: Motor 1 - Front right motor, counter-clockwise
Output 6: Motor 2 - Rear left motor, counter-clockwise
Output 7: Motor 3 - Front left motor, clockwise
Output 8: Motor 4 - Rear right motor, clockwise

So to put your quad motors on outputs 9 to 12 (the auxiliary channels on a Pixhawk) you would use these settings in the advanced parameter list:

SERVO9_FUNCTION = 33
SERVO10_FUNCTION = 34
SERVO11_FUNCTION = 35
SERVO12_FUNCTION = 36

Configuration - Servo Output

This is the assignment of all channels that you can set at the beginning.



MAIN QUADPLANE PARAMETERS

Q_ENABLE = 1

Next, you need to set the parameter Q_ENABLE to 1. This enables all options related to quadplane support. You need to click ‚write' and then ‚refresh' to obtain the full list of parameters.

Q_FRAME_CLASS = 1

Next, set Q_FRAME_CLASS to 1, indicating the selection of the quadplane layout

Q_FRAME_TYPE = 1

This parameter specifies the choice of frame. 1 corresponds to our X frame layout.

Q_M_PWM_TYPE = 0

Specifies the PWM type for the motors. If you are using the recommended ESCs, they support ONESHOT125, and the value of the parameter should be set to 2. If you are using other ESCs, adjust this parameter accordingly. Without specifying it, manual ESC calibration will be required.

Configuration - MAIN QUADPLANE PARAMETERS

Q_M_BAT_VOLT_MAX = 25.2
Q_M_BAT_VOLT_MIN = 19.8

These parameters determines the maximum and minimum battery voltage values. When using 4S packs, you should enter the above values. This allows compensation of throttle in hover mode depending on the current battery state. This way, the throttle position should always provide the same thrust regardless of the voltage.

Q_M_BAT_VOLT_MAX: Battery voltage compensation maximum voltage

Note: This parameter is for advanced users

Battery voltage compensation maximum voltage (voltage above this will have no additional scaling effect on thrust). Recommend 4.2 * cell count, 0 = Disabled

Range	Units
6 to 53	volt

Q_M_BAT_VOLT_MIN: Battery voltage compensation minimum voltage

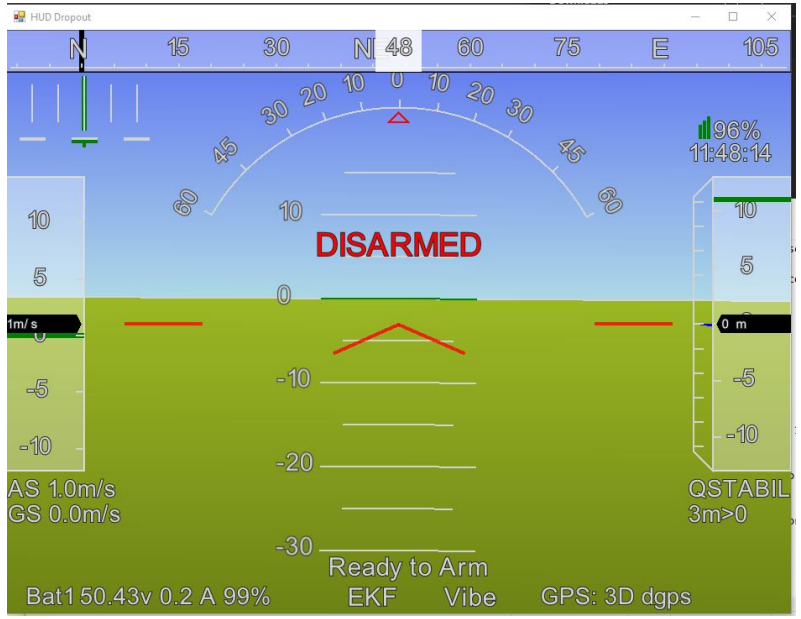
Note: This parameter is for advanced users

Battery voltage compensation minimum voltage (voltage below this will have no additional scaling effect on thrust). Recommend 3.3 * cell count, 0 = Disabled

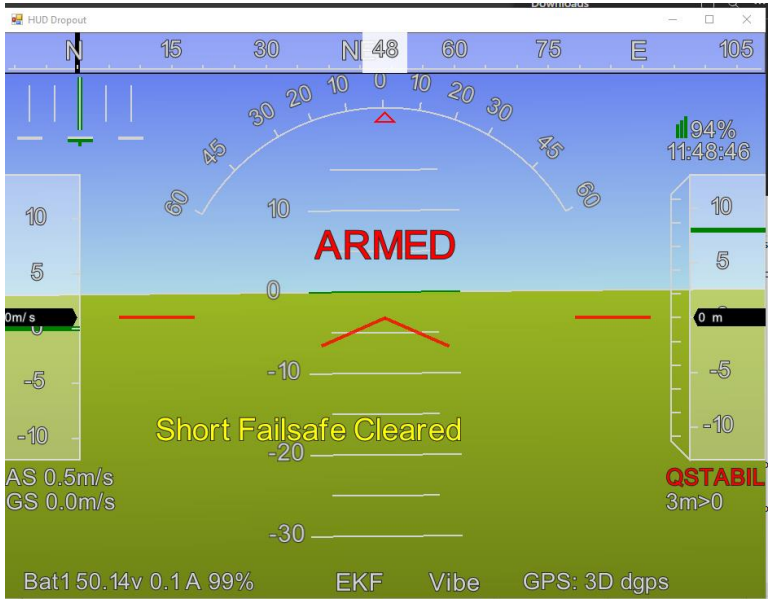
Range	Units
6 to 42	volt

Logs Review – Telemetry Logs

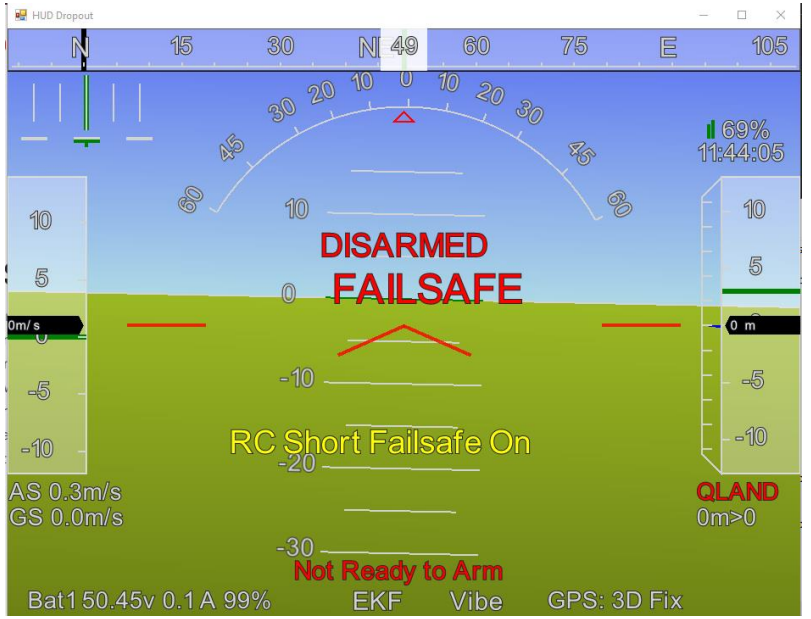
RC Short Failsafe: switched to QLAND



Everything Is normal on qstabilise mode until arming

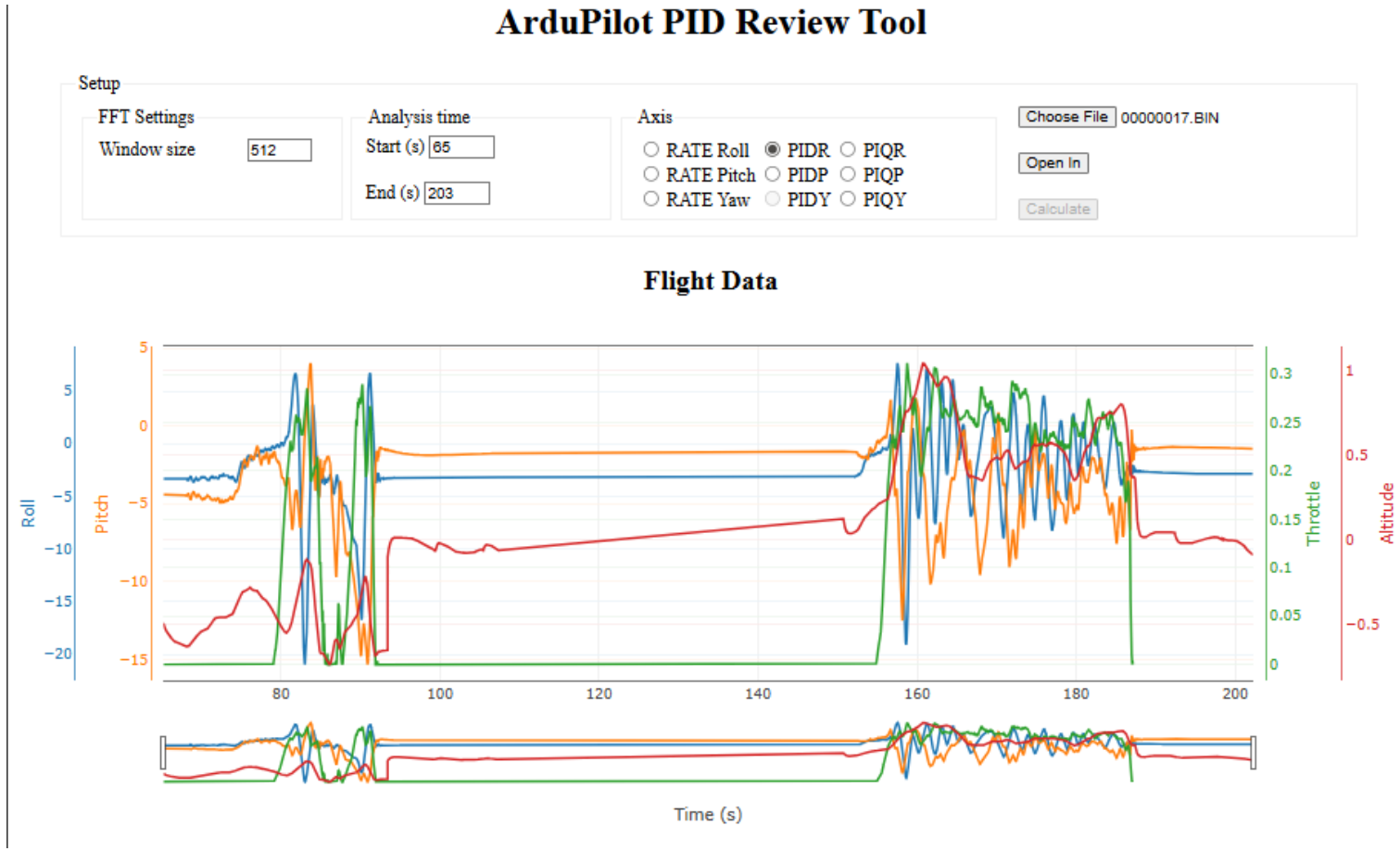


Unknown Error : RC Failsafe



Cause Qland mode : not armable

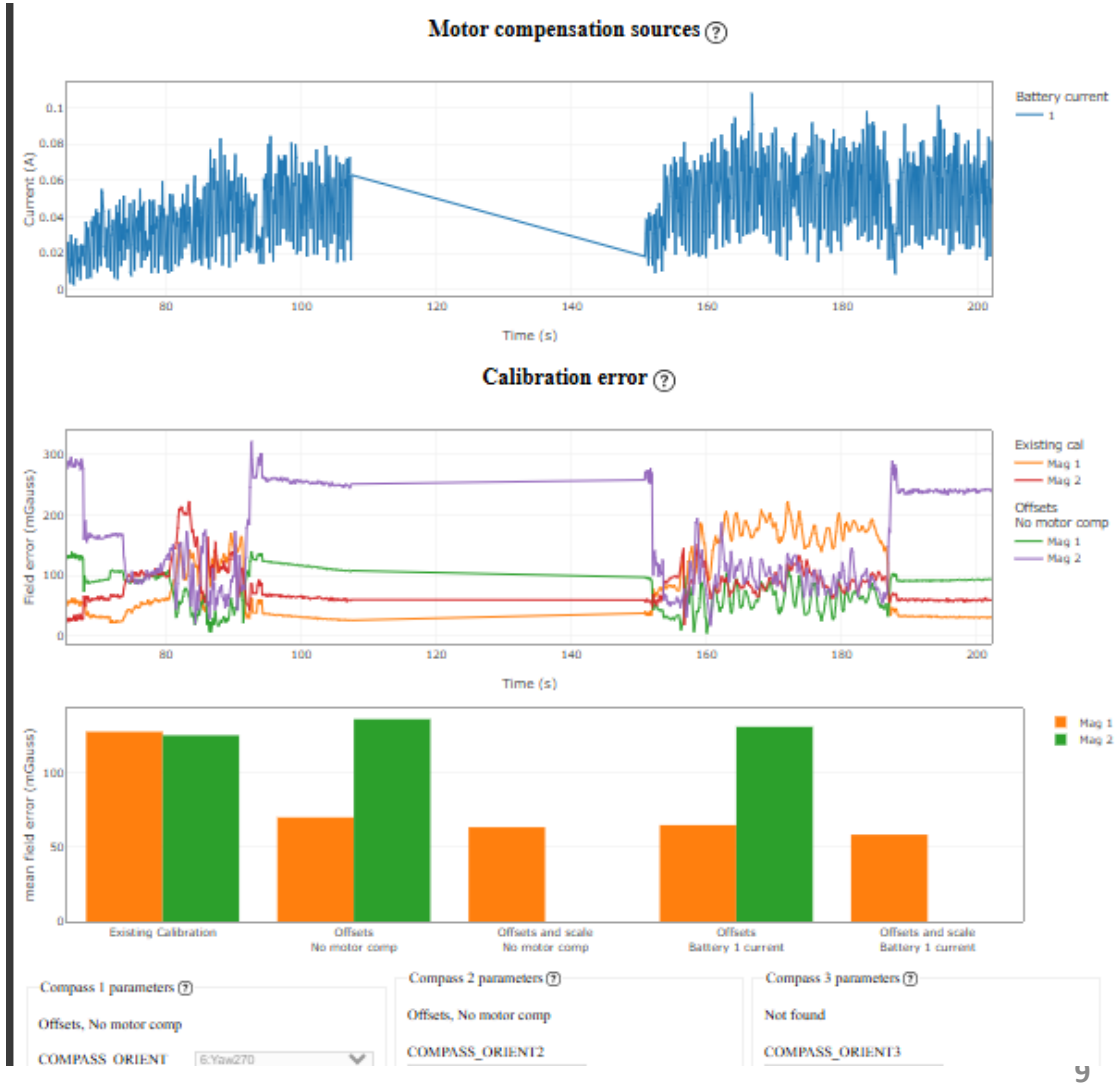
PID Review Tool



ArduPilot MAGFit in flight compass calibration

Expected vs measured body frame magnetic field





COMPASS_ORIENT = yaw270

Video Flight tests review

GX010097





The drone was a bit stable until it landed slowly !

Why the pilot didn’t want to gain more altitude when he saw it abit responsive ?

Video Flight tests review

GX010110



What was the pilot experiencing when the drone was pitching back ?

Video Flight tests review

GX010103





The drone was a bit stable until it landed slowly !

Is the drone land by itself or the pilot

Video Flight tests review

GX010098



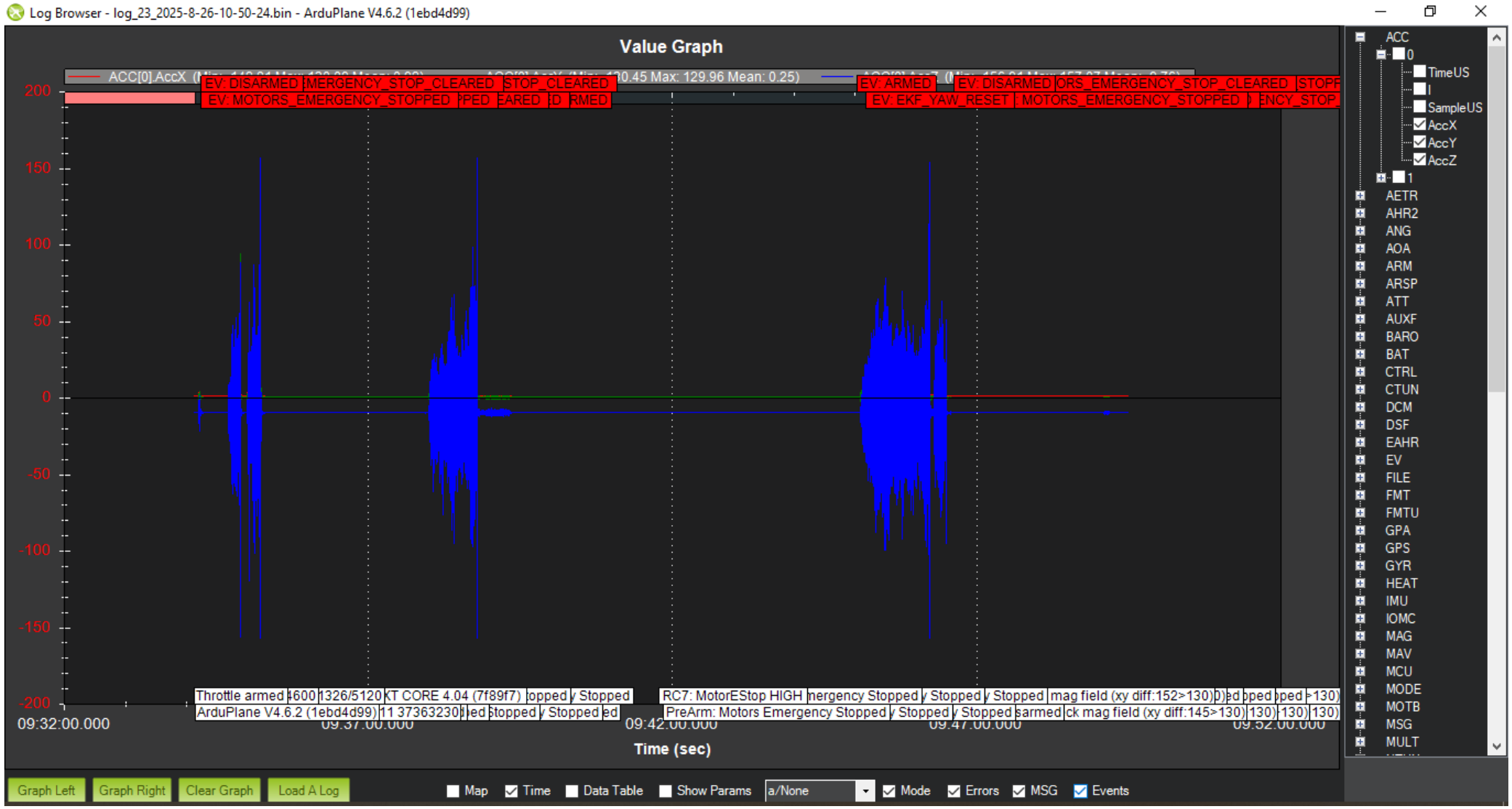


The drone calibrating itself but in instant it oscillating a lot

What the pilot said about the behavior before landing ?

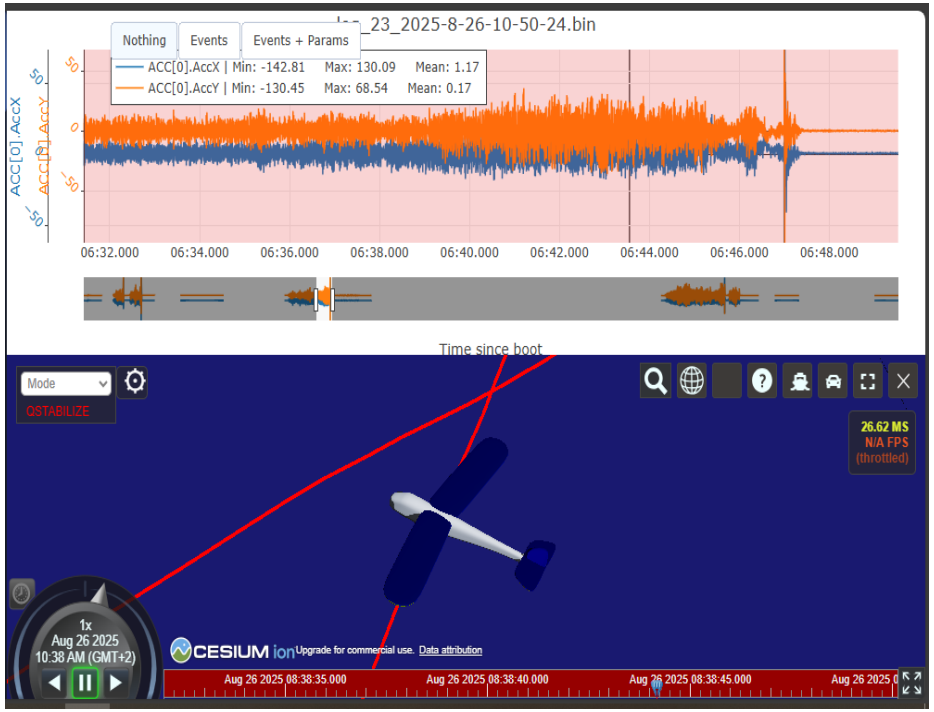
Flight Logs review

Current file: log_23_2025-8-26-10-50-24.bi



Flight Logs review

Current file: log_23_2025-8-26-10-50-24.bin



ACC[0] AccX:

- ✓ Min: **−39.74**, Max: **67.02**, Mean: **0.95**
- ✓ Peak-to-peak ≈ **106.8 m/s²** (already very high).

ACC[0] AccY:

- ✓ First sample: Min: **−69.29**, Max: **46.62**, Mean: **0.23** → Peak-to-peak ≈ **115.9 m/s²** (large).
- ✓ Second sample: Min: **−130.45**, Max: **129.96**, Mean: **0.25** → Peak-to-peak ≈ **260.4 m/s²** (extremely large).

ACC[0] AccZ:

- ✓ Min: **−154.63**, Max: **150.94**, Mean: **−9.77**
- ✓ Peak-to-peak ≈ **305.6 m/s²** (catastrophic).

GYR[0] Gyros:

- ✓ GyrX: (−7.98 to 7.05), GyrY: (−2.73 to 6.46), GyrZ: (−0.33 to 0.47).
  → These are small values, showing **gyros are relatively quiet** compared to accelerometers.

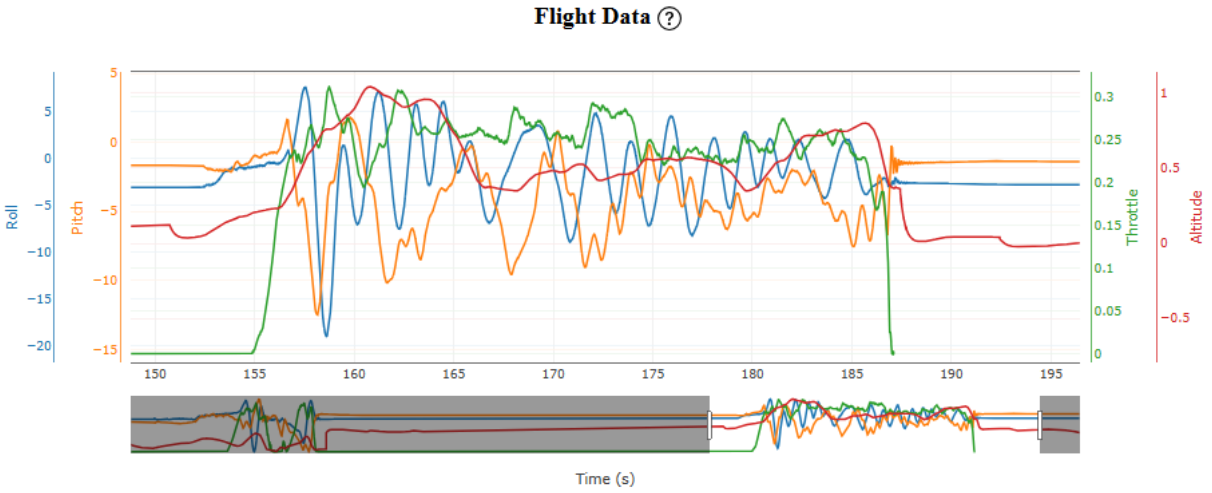
•**Accelerometer extremes (Y & Z axes) are abnormally large**, far beyond what's expected in hover/normal flight.

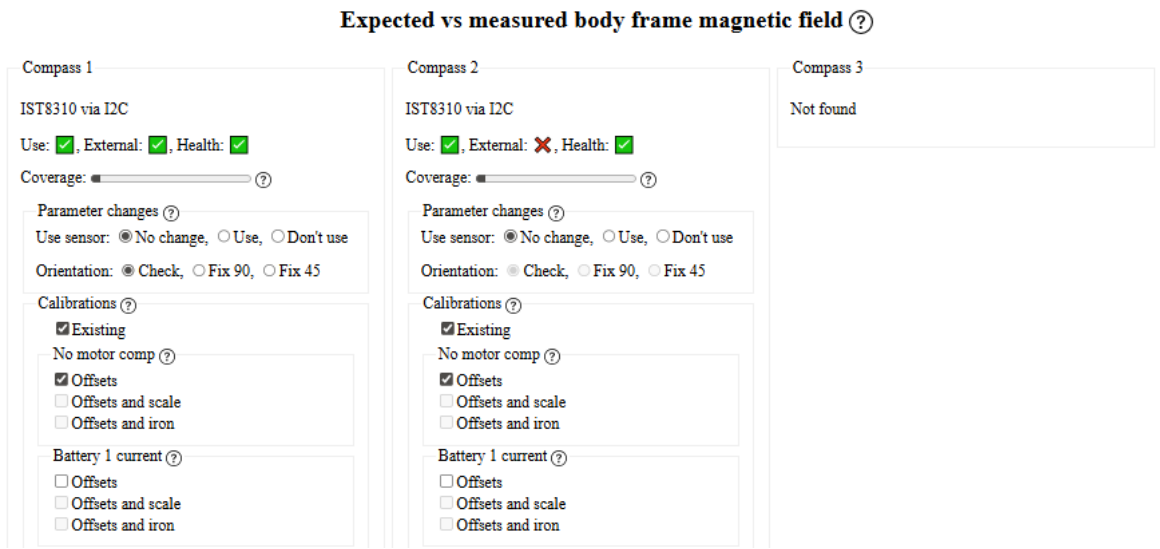
•**High Clip counts** are consistent with the accelerometer being saturated.

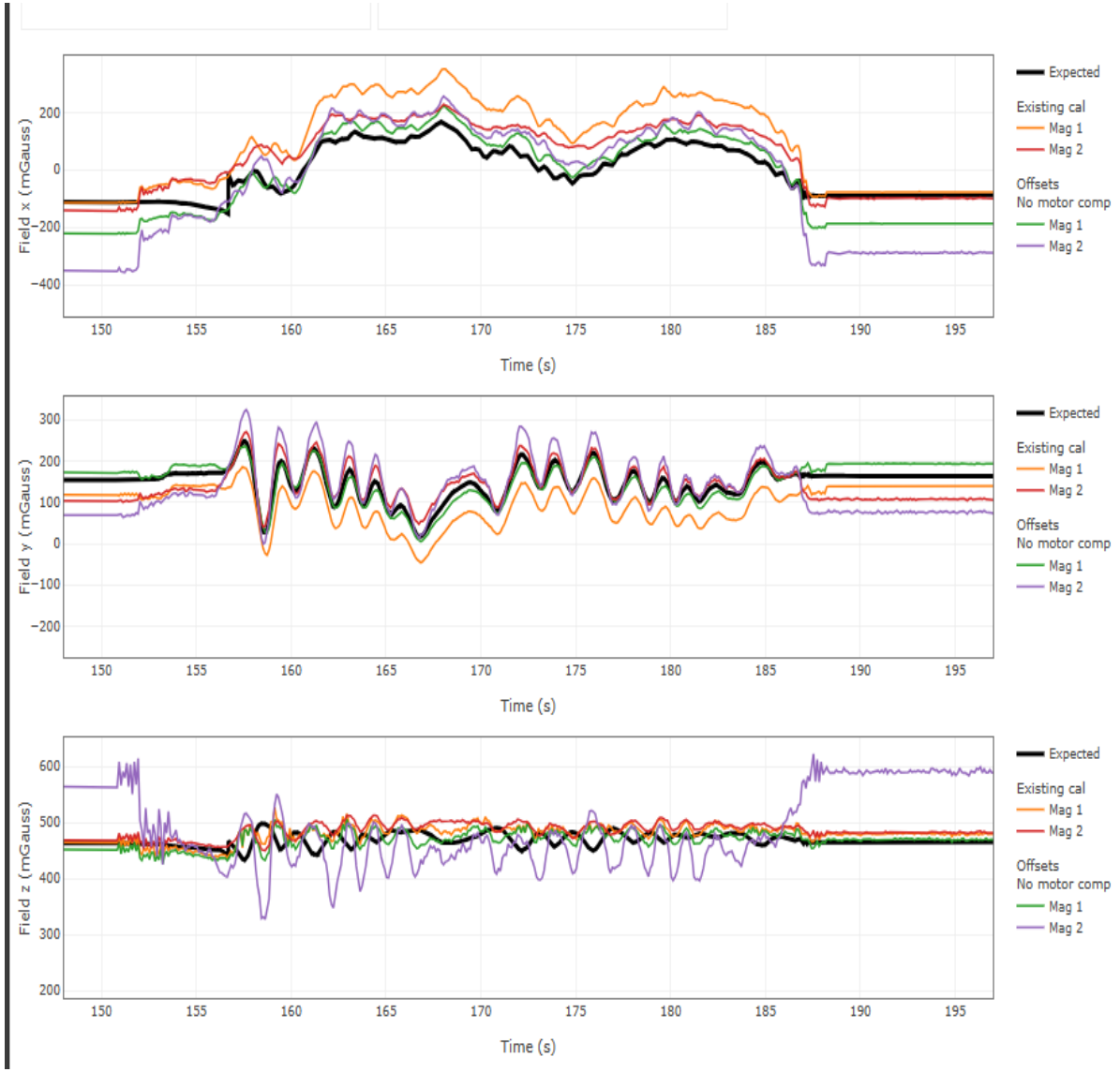
•The **quiet gyros** show this isn't smooth vibration — it's impulsive shocks (props, shafts, or something slamming the FC).

ArduPilot MAGFit in flight compass calibration

Expected vs measured body frame magnetic field



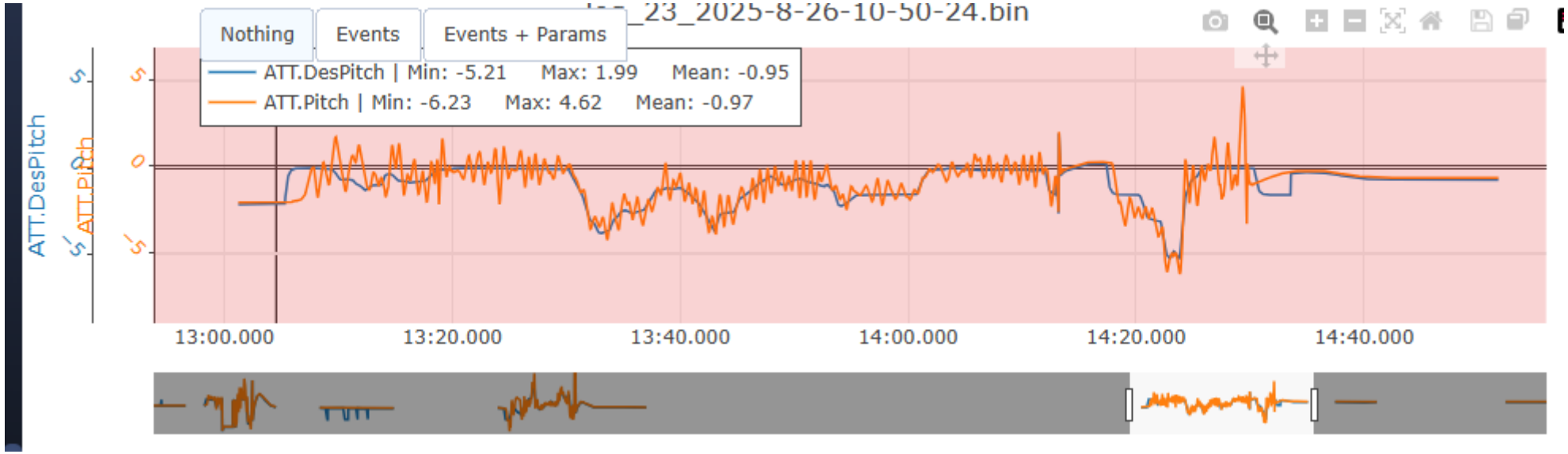




PID Review Tool

Pitch : ATT.DesPitch (blue) vs ATT.Pitch (orange)

The orange (actual pitch) does follow the blue (desired pitch), but it is **noisy and wobbly**



Stick Inputs are Small
Desired pitch (blue) only moves about ±2°.
But the actual pitch (orange) swings **much further** (up to ±6°).
This confirms **the controller is overreacting** to even small inputs.

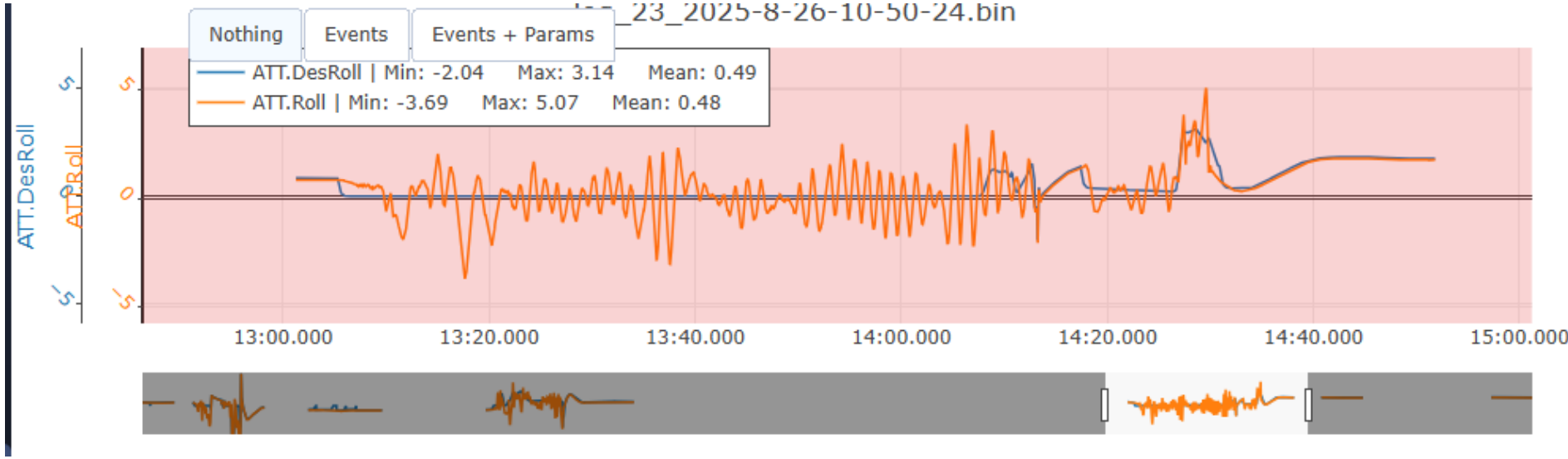
Overshoot & Oscillation
- Similar to roll, the pitch overshoots the command and rings before settling.
- That means **Pitch P-gain is too high**.

Tracking
- The orange (actual pitch) does follow the blue (desired pitch), but it is **noisy and wobbly**.
- Especially after ~14:20:00, the orange line overshoots and oscillates.

PID Review Tool

Roll : ATT.DesPitch (blue) vs ATT.Pitch (orange)

Aggressive oscillations (orange line)
Your **ANG.Roll** is very “jagged”, meaning the controller is fighting too hard → **P-term too high**.



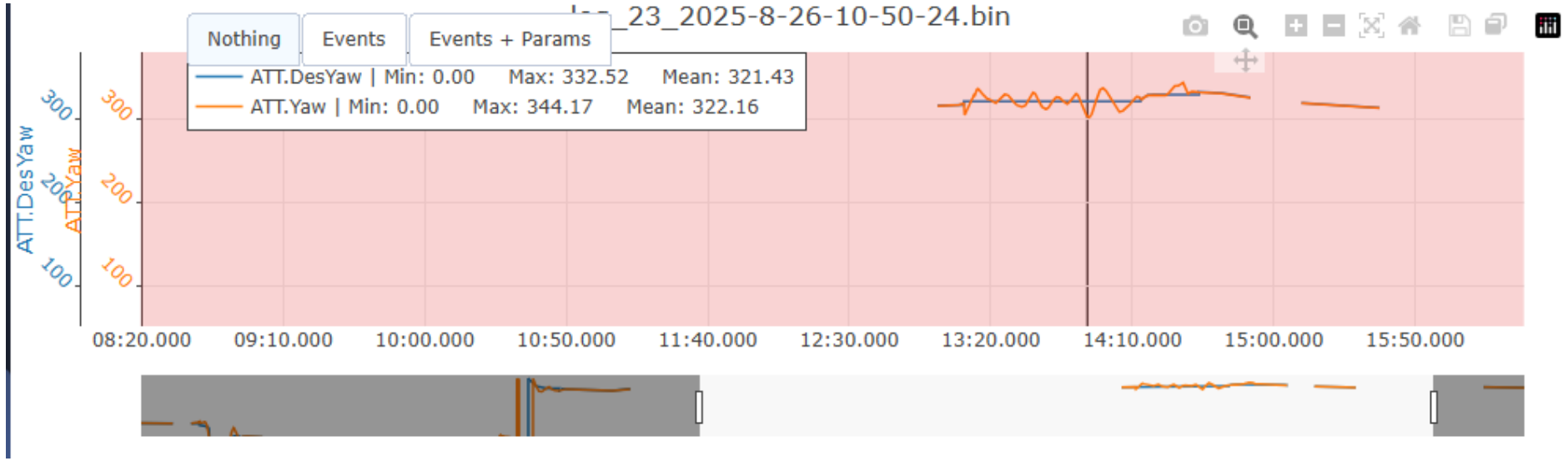
- Reduce **Q_A_RAT_RLL_P** by ~40%.
- 0.20 → set to 0.12.

Overshoot
When you command a roll (blue line steps up), the actual roll (red) **overshoots strongly** and then oscillates before settling.
This is a textbook case of **too much RATE_P** or too much ANG_P gain.

Des vs Actual
Desired is only about ±3° roll, but the actual oscillates much more violently.
That means the **flight controller is amplifying your small stick inputs.**

PID Review Tool

Yaw : ATT.DesYaw vs ATT.Yaw



Alignment

When DesYaw moves, Actual Yaw tracks it **without major delay or overshoot**.
This means **Yaw PID is okay**.

Overshoot & Oscillation

- Similar to roll, the pitch overshoots the command and rings before settling.
- That means **Do NOT touch yaw PIDs for now** (Q_A_RAT_YAW_P, I, D)..

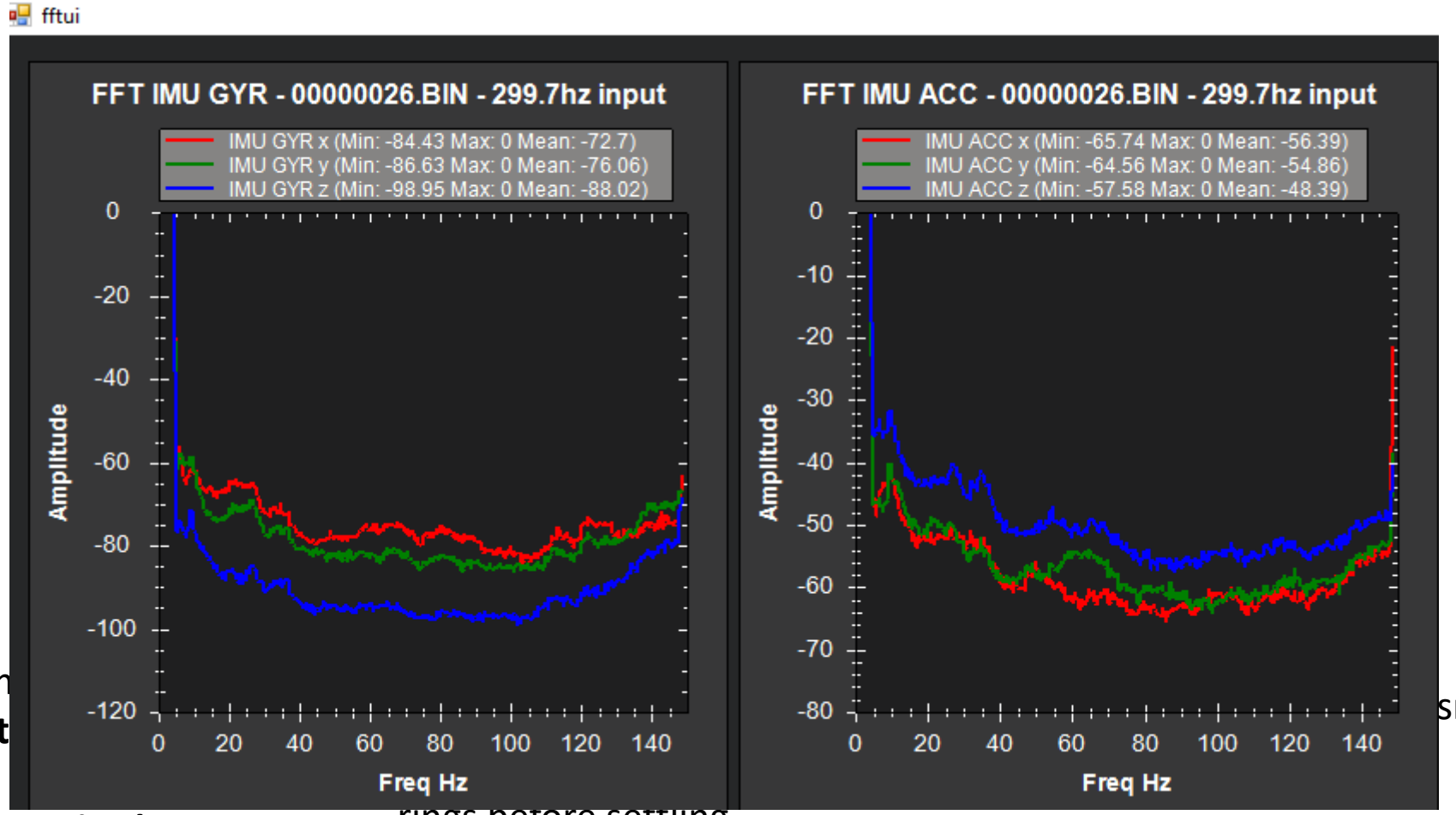
Actual Yaw (orange)

It's fairly stable, only small oscillations (~1–2°).

No big overshoot or hunting like Roll/Pitch.

FFT

Setup : Advanced -> FFT(plot a FFT from log)



Alignment

When DesYaw m

tracks it **without overshoot**.

This means **Yaw PID is okay**.

small

rings before settling.

- That means **Do NOT touch yaw PIDs for now** (Q_A_RAT_YAW_P, I, D)..

No big overshoot or hunting like Roll/Pitch.

What we can do

We can perform a test to see if the motors interfere the compass signals



- You should hear your ESCs arming beep
- Raise the throttle slowly to between 50% ~ 75% (the props will spin!) for 5 ~ 10 seconds
- Quickly bring the throttle back down to zero
- Press the **Finish** button to complete the calibration
- Check the % of interference displayed. If it is less than 30% then your compass interference is acceptable and you should see good Loiter, RTL and AUTO performance. If it is 31% ~ 60% then the interference is in the "grey zone" where it may be ok (some users are fine, some are not). If it is higher than 60% you should try moving your autopilot further up and away from the sources of interference or consider purchasing an external compass (or GPS+compass module (some of these)).

And also use the gps mount to keep it far away from the high power cables



What we can do

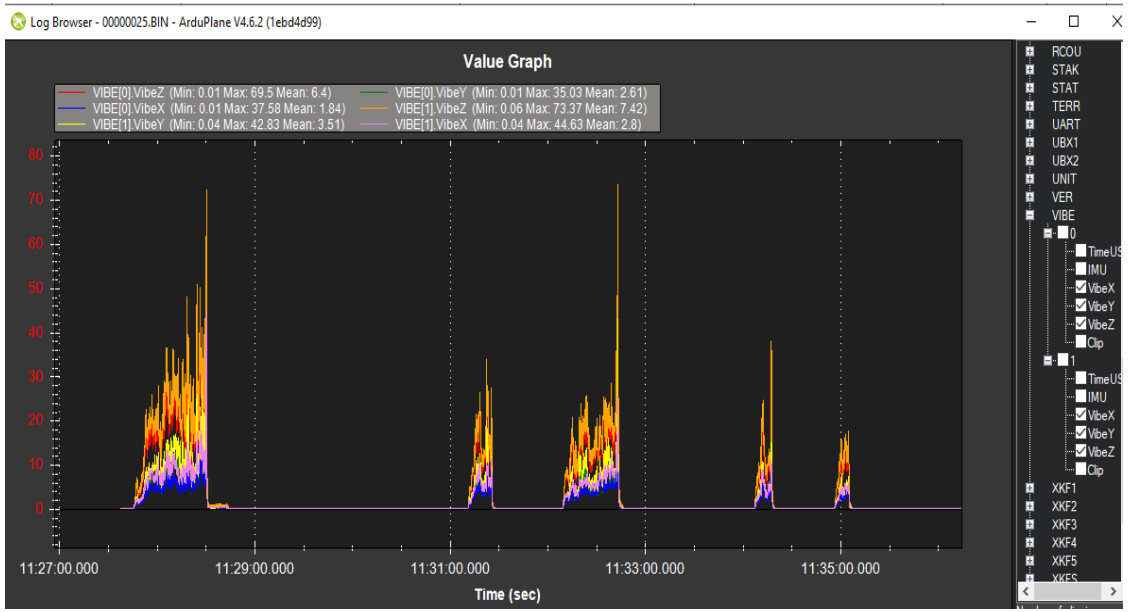
Can we attach the drone with cables somewhere so we can perform the calibrations while the electronics in the drone





Measuring Vibration from Log

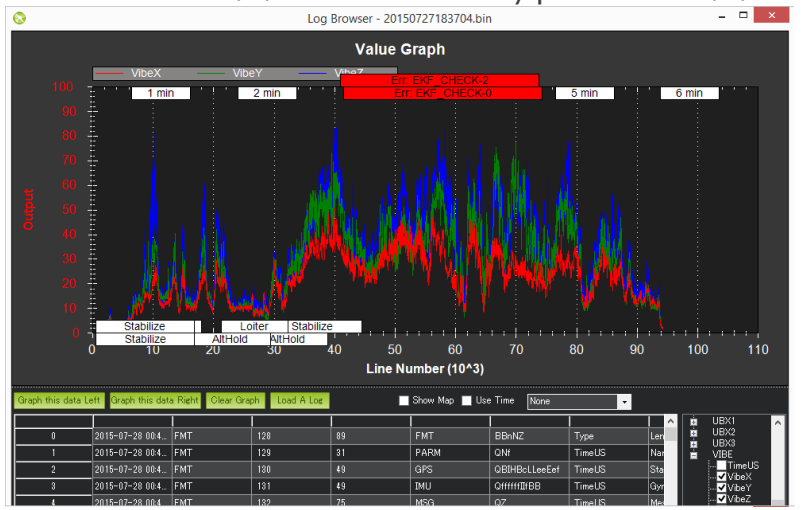
Autopilots have accelerometers that are sensitive to vibrations. These accelerometer values are combined with barometer and GPS data to estimate the vehicle's position





shows normal levels which are below 15m/s/s but occasionally peak to 30m/s/

VIBE[0] VibeZ (Min: 0.01 Max: 69.5 Mean: 5.53)
VIBE[0] VibeY (Min: 0.01 Max: 35.03 Mean: 2.26)
VIBE[0] VibeX (Min: 0.01 Max: 37.58 Mean: 1.59)
VIBE[1] VibeZ (Min: 0.06 Max: 73.37 Mean: 6.42)
VIBE[1] VibeY (Min: 0.04 Max: 42.83 Mean: 3.04)
VIBE[1] VibeX (Min: 0.04 Max: 44.63 Mean: 2.42)

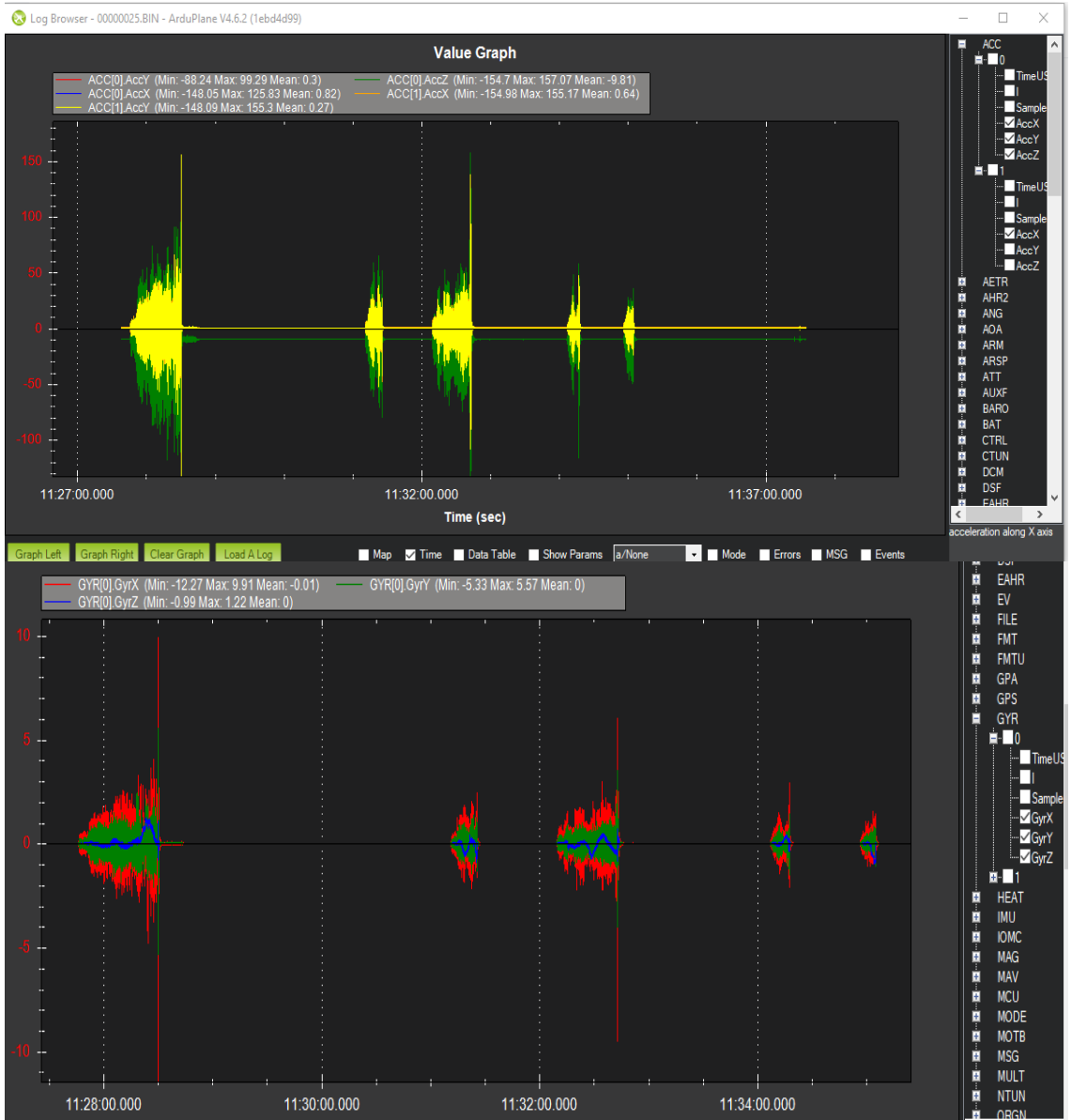


This is an example of a vehicle that had position estimate problems due to high vibrations.

- ✓ **Critical: VibeZ peaks 69.5 & 73.37** very high impulsive shocks.
- ✓ **Other axes:** VibeX/VibeY peaks up to ~44; means all axes see strong vibration, Z worst.
- ✓ **Interpretation:** not normal hover vibration, likely **prop/prop-mount issues, motor shaft/bearing play, or hard impacts/transmitted shocks to FC** (Z axis dominated). Gyro quietness from previous logs + high VIBE = impulsive shocks

Measuring Vibration from Log

Check the scale on the left and ensure that your vibration levels for the **AccX and AccY are between -3 and +3**. For **AccZ the acceptable range is -15 to -5**.





- Check the scale on the left and ensure that your vibration levels for the AccX and AccY are between -3 and +3. For AccZ the acceptable range is -15 to -5. If it is very close or over these limits you should refer back to the Vibration Damping page for possible solutions.

ACC[0].AccY (Min: -88.24 Max: 99.29 Mean: 0.3)
ACC[0].AccX (Min:-148.05 Max: 125.83 Mean: 0.82)
ACC[1] AccY (Min:-148.09 Max: 155.3 Mean: 0.27)
ACC[0].AccZ (Min:-154.7 Max: 157.07 Mean: -9.81)
ACC[1] AccX (Min-154.98 Max: 155.17 Mean: 0.64).

Interpretation : these are **catastrophic shock magnitudes** on all axes; two accel sensors show similar large p–p → whole-airframe mechanical/prop/motor problem (not a single bad IMU).

GYR[0].GyrX (Min:-12.27 Max: 9.91 Mean: -0.01)
GYR[0] GyrZ (Min: -0.99 Max: 1.22 Mean: 0)
GYR[0] GyrY (Min: -5.33 Max: 5.57 Mean: 0) [1]

Setup the “First Notch Filter”

Goal : Filter out the lift-motor vibration peak (and its harmonics) so the attitude controller isn’t fighting noise; especially in Q-modes.

(mechanical fixes → logging → FFT → set notches → verify → tune)

Quick summary (what we’ll do)

1.Fix the obvious mechanical vibration sources first (props, motors, mountings).

2.Collect high-rate IMU data (raw IMU or IMU batch sampling) while the aircraft hovers.

3.Use ArduPilot’s Filter/FFT tools to find the dominant gyro frequency and bandwidth.

4.Enable and configure the **Harmonic Notch** (INS_HNTCH_*) using the measured values.

5.Re-test, verify the FFT / Vibe graphs, then relax or raise gyro low-pass (INS_GYRO_FILTER) and finish PID tuning.

(ArduPilot docs with full details: see the Notch/QuadPlane tuning pages below.) Managing Gyro Noise with the Dynamic Harmonic Notch Filters — Plane documentation

Setup the “First Notch Filter”

Mechanical checklist (do these first)

These are the most common causes. Fixing them is faster and safer than trying to over-filter in software.



1. **Tighten** all prop nuts, motor screws and motor mounts.
2. **Balance props** (static balancing at minimum; dynamic if possible).
   Badly unbalanced big props = lots of aliased noise.

1. **Check motor bearings/shafts** for play or wobble. Replace suspect motors.

2. **Stiffen the frame / booms** (remove flexy arms, add braces). X-frames can twist under thrust check for twist under load.



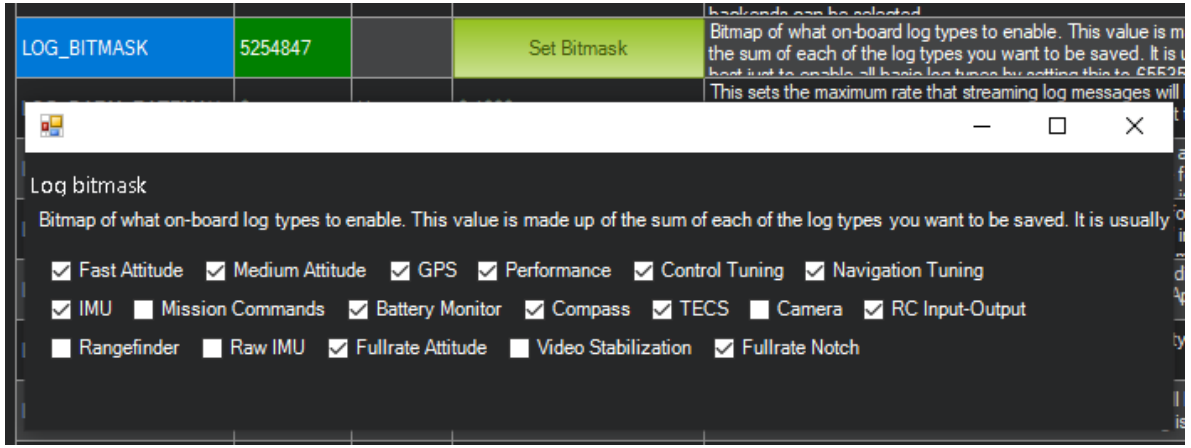
3. **Flight controller mounting**: use a recommended mount or short soft standoffs, but not an overly floppy mount that introduces resonance. If you move the FC, update IMU position params **(INS_POS*).**

4. **Route power wires** and heavy wiring away from the FC/IMU and GPS.
   (If vibration remains high after mechanical work, proceed to logging + notch filters.)
   ArduPilot.org

Setup the “First Notch Filter”

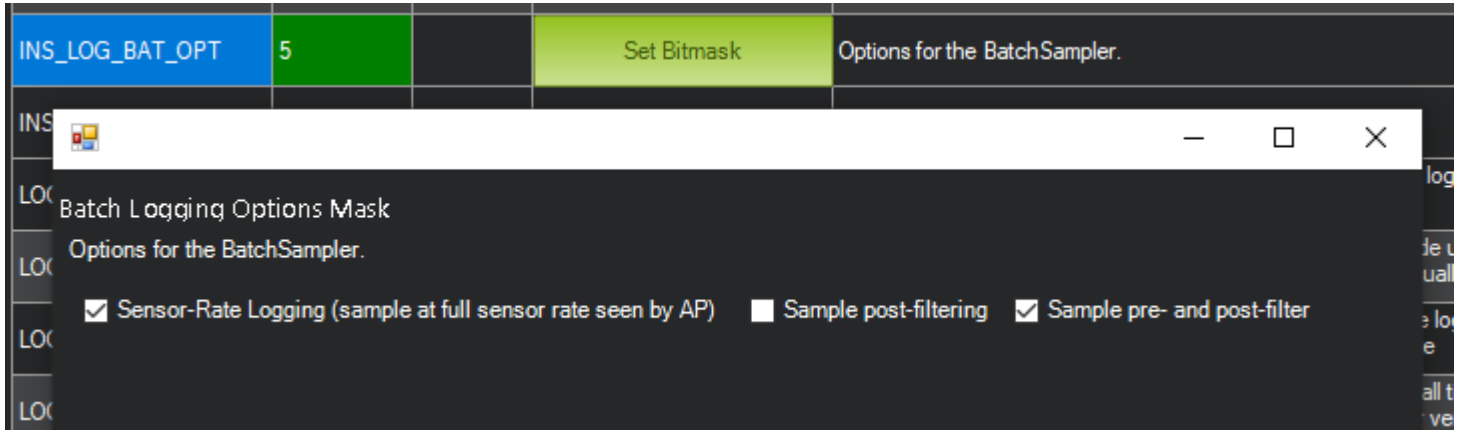
1: Setup - Logging

FFT Setup page -> **Sample count = 2048**



LOG_BITMASK = 5254847

We must capture pre-filter IMU data so you can identify the motor/prop frequency.



INS_LOG_BAT_OPT 5

INS_LOG_BAT_MASK = 1 (log IMU1)

What we can do

1: Setup – Harmonic Notches FFT

Setup the “First Notch Filter”

Goal : Filter out the lift-motor vibration peak (and its harmonics) so the attitude controller isn’t fighting noise; especially in Q-modes.

INS_HNTCH_ENABLE = 1 # turn on harmonic notch → reboot.
INS_HNTCH_MODE = 4 # 1=Throttle tracking (works without RPM/ESC data)
INS_HNTCH_HMNCS = 7 # fundamental + 2 harmonics
INS_HNTCH_ATT = 40 # attenuation (30–50 typical)
INS_HNTCH_REF = 1 no scaling required

Initial frequency guess for heavy lift quads (18–22" props typically hover ~60–100 Hz)
INS_HNTCH_FREQ = FFT_MINHZ = 20 Hz
INS_HNTCH_BW = INS_HNTCH_FREQ /2 = 10hz
INS_HNTCH_FM_RAT = 1 prevent the notch going below INS_HNTCH_FREQ
Logging (helps you verify & fine-tune)
LOG_BITMASK = 131071 # or ensure IMU raw/fast is enabled in your GCS

INS_GYRO_FILTER = 20 # Safe for 25" props
INS_ACCEL_FILTER = 10
Q_A_RAT_RLL_FLTD = 10
Q_A_RAT_PIT_FLTD = 10

Static notch for frame resonance

INS_NOTCH_ENABLE = 1
INS_NOTCH_FREQ1 = <frame frequency from FFT, e.g. 40>
INS_NOTCH_BW1 = 8–12 (narrow)
INS_NOTCH_ATT1 = 18–25
Write params and reboot.

What we can do

1: Setup – Harmonic Notches - Throttle

Setup the “First Notch Filter”

Goal : Filter out the lift-motor vibration peak (and its harmonics) so the attitude controller isn’t fighting noise; especially in Q-modes.

INS_HNTCH_ENABLE = 1 # turn on harmonic notch
INS_HNTCH_MODE = 1 # 1=Throttle tracking (works without RPM/ESC data)
INS_HNTCH_HMNCS = 4 # fundamental + 2 harmonics
INS_HNTCH_ATT = 40 # attenuation (30–50 typical)
INS_HNTCH_REF = Q_M_THST_HOVER # set equal to your hover throttle (e.g. 0.30)

INS_HNTCH_FREQ = 80 # Starting center frequency (Hz)
INS_HNTCH_BW = INS_HNTCH_FREQ /2 = 40hz # Bandwidth (half of freq)

INS_HNTCH_FM_RAT = 0.7 # Tracks down to ~70% of hover
INS_HNTCH_OPTS = 0 # Leave advanced options off
Logging (helps you verify & fine-tune)
LOG_BITMASK = 131071 # or ensure IMU raw/fast is enabled in your GCS

if Throttle based make sure your motor voltage compensation is setup (see Q_M_BATT_x params for QuadPlane)

(Throttle-Based Harmonic Notch) vs (In-Flight FFT)

INS_HNTCH_MODE = 1 (Throttle-Based Harmonic Notch):

- **How it works:** This mode scales the notch filter's center frequency based on the main throttle output. It assumes a relationship between throttle percentage and motor RPM/noise frequency.

•Pros: Easy to set up, no additional hardware required.
•Cons: Less accurate than RPM-based or FFT-based methods as the relationship between throttle and motor frequency isn't perfectly linear or constant across different flight conditions. It's an estimation.

INS_HNTCH_MODE = 4 (In-Flight FFT Based Harmonic Notch):

- **How it works:** The autopilot performs a Fast Fourier Transform (FFT) on the raw gyro data in real-time to identify the dominant noise frequencies and then places the notch filters dynamically based on these findings.

•Pros: Highly accurate as it directly measures noise frequencies. Works well on H7 boards like your Pixhawk 6C which have the processing power.
•Cons: More CPU intensive than throttle-based. Requires FFT_ENABLE = 1 and setting up FFT_MINHZ, FFT_MAXHZ, and FFT_WINDOW_SIZE correctly.

Recommendation for Pixhawk 6C (H7):
For the Pixhawk 6C with an H7 processor, the recommendation is INS_HNTCH_MODE = 4 (In-Flight FFT). This method provides better performance and more accurate noise suppression compared to the throttle-based method, because of the flight controller's power.

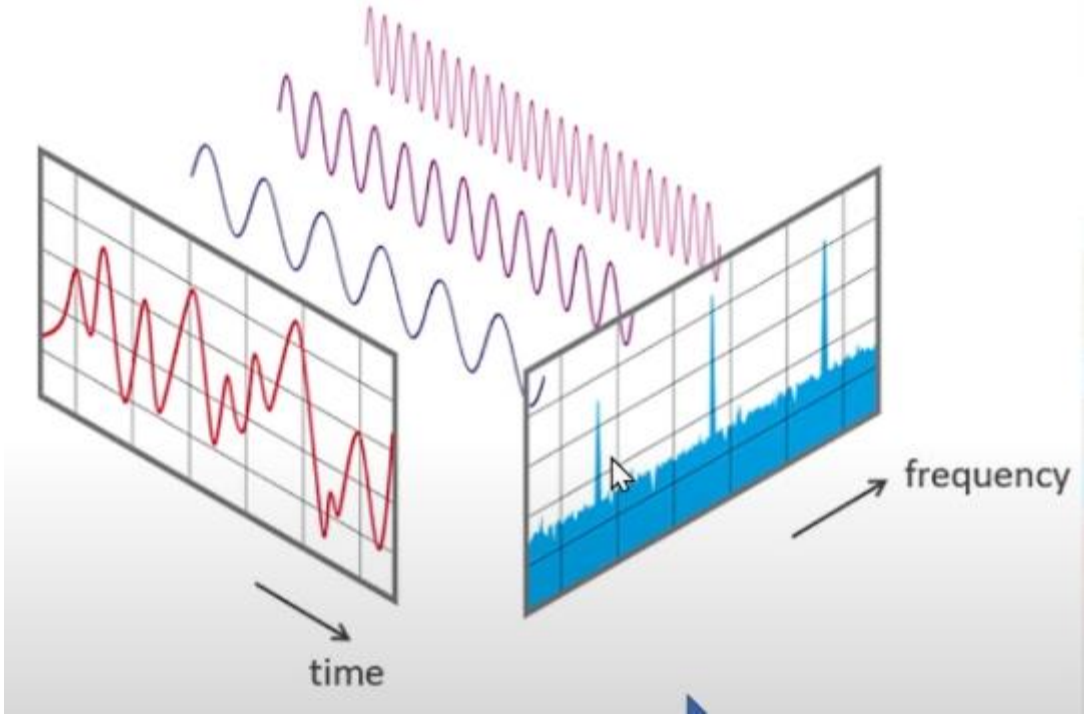
Setup the “First Notch Filter”

1: Setup – Dynamic FFT (Fast Fourier Transform)

Set **FFT_ENABLE = 1** to enable FFT and show parameters
Set **INS_HNTCH_MODE = 4** to use FFT to drive the notches
INS_HNTCH_REF: Set to 1
Set **FFT_MAXHZ** = X
X = 0.8*Vbatt*KV/60
Vbatt=50.4 V
KV=180 (from T-Motor AT8020 KV180)

X = 0.8*50.4*180/60
Y = 500 / 25

Set **FFT_MINHZ** = Y
Y = 500 / Dprop
Dprop = Prop Diameter in Inches = 25
FFT_MAXHZ = 120.96 Hz
FFT_MINHZ = 20 Hz
FFT_NUM_FRAMES = 0 FFT needs to be reactive to changing RPM
FFT_OPTIONS = 1 to apply low pass filters before doing the FFT
FFT_WINDOW = 1024 if CPU load allows or as high as possible.



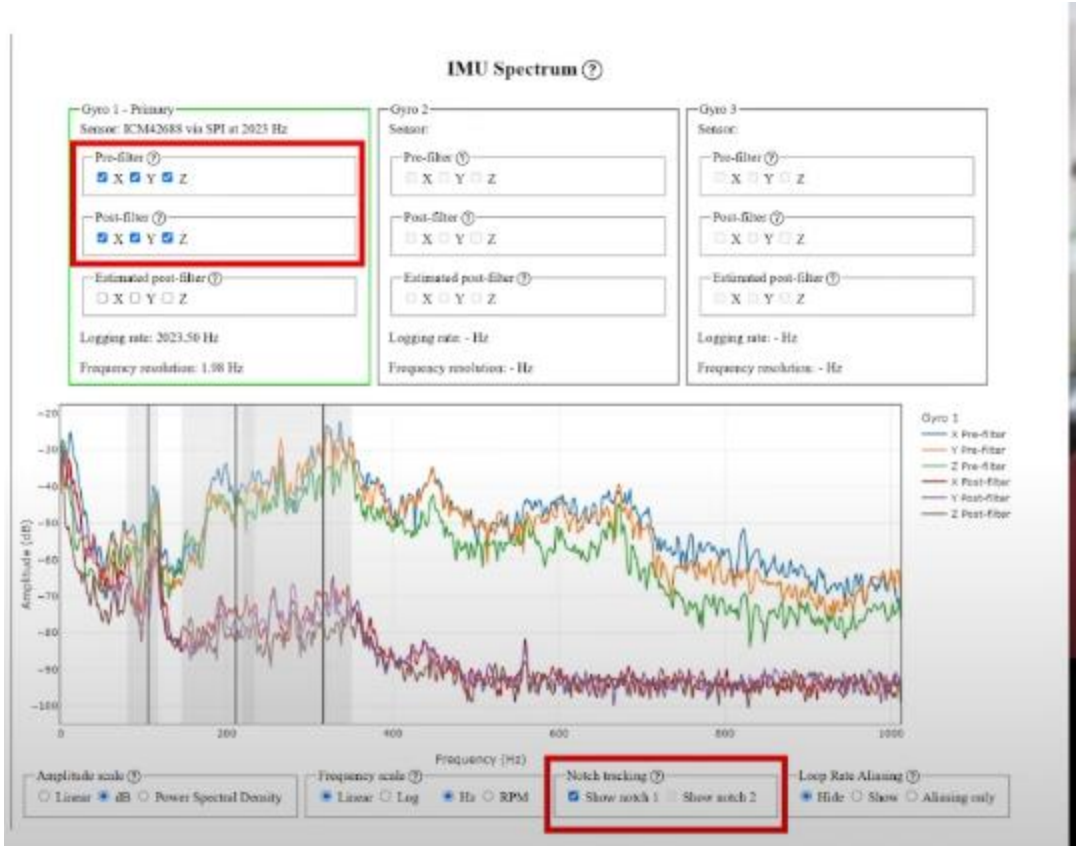
Setting FFT_WINDOW_SIZE for Pixhawk 6C (H7 Processor),
you should set **FFT_WINDOW_SIZE = 512** (If your Pixhawk 6C's CPU speed allows, you will benefit from FFT_WINDOW_SIZE = 1024)

Checking – Filter Review tool

1: Setup – IMU Spectrum

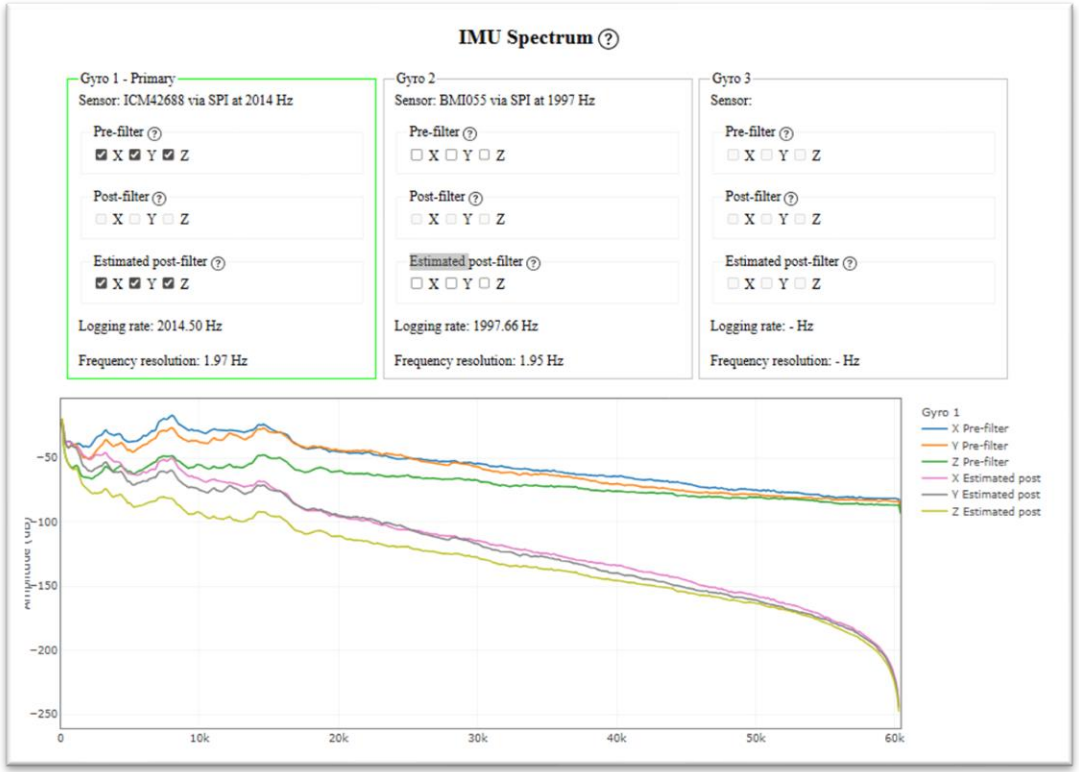
With the logging settings we should be able to see

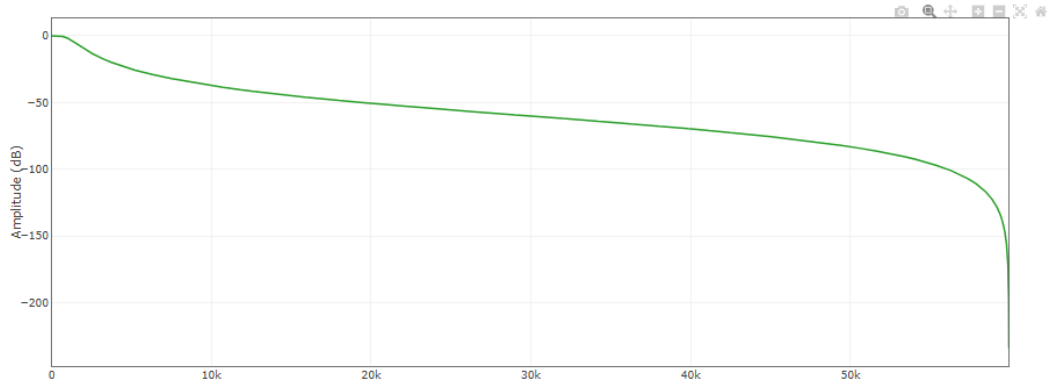
- Pre-Filter Gyro Data
- Post-Filter Gyro Data
- Notch Tracking



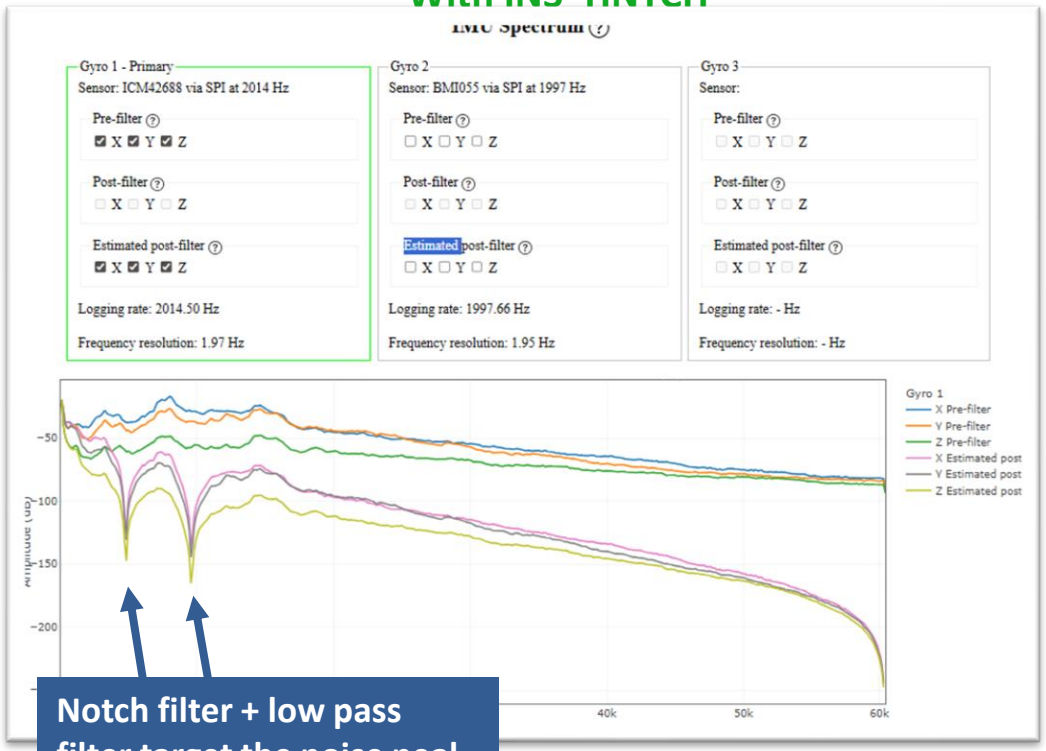
To understand Harmonic Notch Filter better

Without INS_HNTCH_

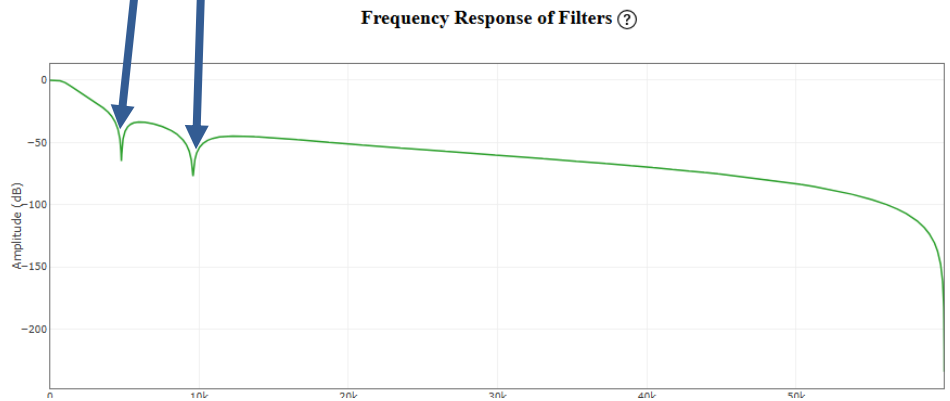




With INS HNTCH



Notch filter + low pass filter target the noise peal



Questions for Mr.Marc

T-Motor AT8020

AT8020 KV180	25*10	40%	44.68	11.61	518.70	2970	1.140	3277	6.30	93℃
		45%	44.31	15.40	682.40	3330	1.440	4226	6.20	
		50%	44.35	19.36	858.60	3600	1.720	5154	6.00	
		55%	44.16	24.25	1070.90	3870	2.010	6120	5.70	
		60%	44.07	30.09	1326.10	4170	2.370	7307	5.50	
		65%	43.72	37.41	1635.60	4500	2.810	8699	5.30	
		70%	43.77	45.69	1999.90	4830	3.230	10085	5.00	
		75%	43.26	53.95	2333.90	5100	3.620	11455	4.90	
		80%	43.17	63.48	2740.40	5400	4.030	12741	4.60	
		90%	42.40	84.18	3569.20	5880	4.890	15371	4.30	
		100%	41.53	111.25	4620.20	6330	5.800	18121	3.90	





Test flight & verification

1. Hover in a calm area (QHOVER / QSTABILIZE) for ~30–120 s with minimal stick input.
2. Download .bin log. Open with Filter Review or Mission Planner's FFT tools and compare pre- and post-filter spectra, the main gyro peak should be greatly reduced. Also check **the Vibe graph in Mission Planner (VibeX, VibeY, VibeZ). Vibe < ~30 m/s/s is normally acceptable; >60 m/s/s usually causes problems**. ArduPilot.org+1
3. Watch CPU load: **PM.Load and PM.NLon** in the log. If CPU load is very high after enabling notches, **reduce the number of notches/harmonics** or disable per-IMU notches. The docs recommend staying under ~60% CPU load for heavy FFT use.